



Université de Sherbrooke
Summer school in biodiversity modelling August 27th, 2026

Trophic metacommunities

Dominique Gravel



PRINCETON
LANDMARKS
IN BIOLOGY

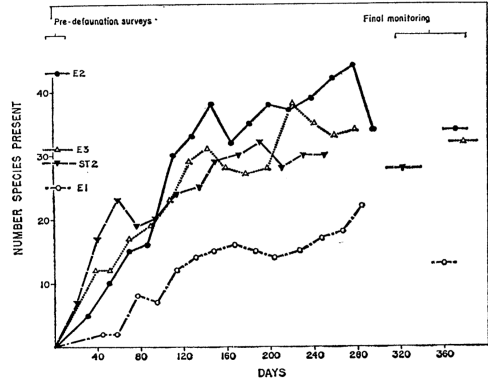
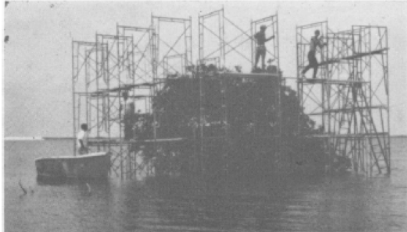
THE THEORY OF
ISLAND
BIOGEOGRAPHY



WITH A NEW PREFACE BY EDWARD O. WILSON

ROBERT H.
MACARTHUR

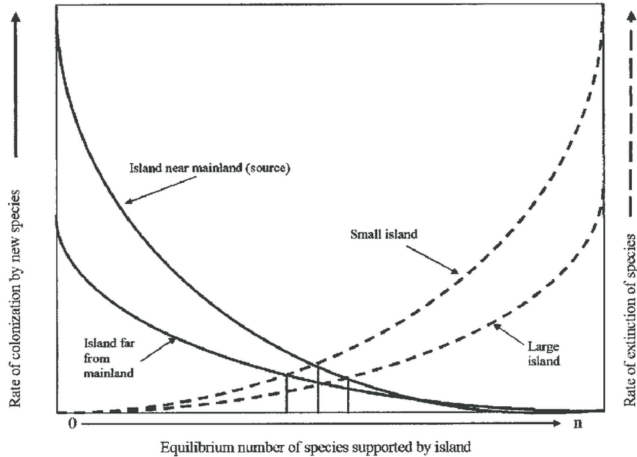
EDWARD O.
WILSON



Simberloff and Wilson, 1967, 1969. Ecology.



Classic IBT Graphical theory





Classic IBT Model

Very similar to the Levins model :

$$\frac{dp}{dt} = c(1 - p) - ep$$

With solution

$$p^* = \frac{c_i}{c + e}$$

And

$$S^* = Pp^* = P \frac{c_i}{c + e}$$



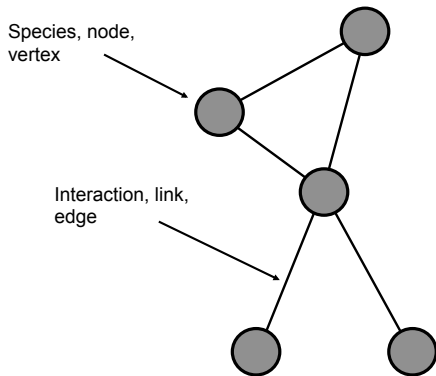
Classic IBT Lessons

- ▶ Importance of colonization and extinction dynamics
- ▶ Local communities are sampled from the regional pool
- ▶ Dynamic equilibrium
- ▶ Hypothesis for species-area relationship
- ▶ Speculations on evolutionary dynamics



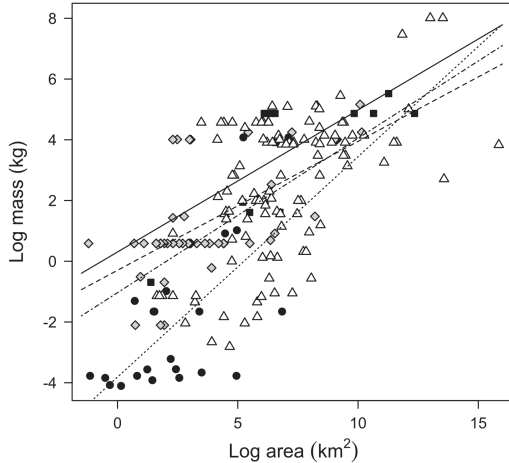
Classic IBT Limits

A community is more than a species list





Classic IBT Limits



Millien and Gonzalez. 2011. JBiog.

Exercise



Classic IBT

Simulate community assembly dynamics on islands without ecological interactions.

Explore the effect of island size and distance to mainland on

- ▶ Time it takes to reach equilibrium
- ▶ Species turnover after equilibrium



Classic IBT Specialized interactions

- ▶ Now consider a regional pool of plants (P) and of herbivores (H)
- ▶ Plants do not have any constraint to colonize islands
- ▶ Herbivore species j requires plant species $i = j$ to be present on the island to colonize
- ▶ Herbivore species j goes extinct if plant species $i = j$ goes extinct

What is the equilibrium species richness of plants and herbivores ?

Predator-prey assembly



Levins model Two trophic levels

Let's first consider a two trophic levels model where the predator requires the presence of the prey to occupy a patch. We have the general equation for trophic level l :

$$\frac{dp_l}{dt} = c_l p_l (h_l - p_l) - e_l p_l$$

where h_l is the fraction of patches suitable for that species, i.-e. the occupancy of the lower trophic level ($h_2 = p_1$).

We call this model "donor-controlled" because it is the lower trophic level that determines the dynamics of higher level, without top-down regulation.



Levins model Two trophic levels

From basic metapopulation theory we know that

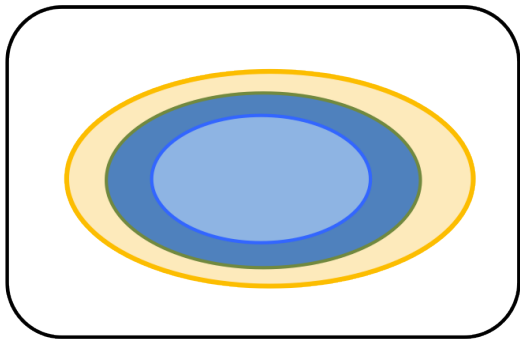
$$p_1^* = 1 - \frac{e_1}{c_1}$$

And solving for the higher trophic level we get

$$p_2^* = h_2 - \frac{e_2}{c_2} = 1 - \frac{e_1}{c_1} - \frac{e_2}{c_2}$$

With this formulation, it is easy to guess that the second trophic level requires that $e_2/c_2 < 1 - e_1/c_1$ for persistence

Concept of spatial inefficiency





Limits to food chain length

Taking this reasoning to multiple trophic levels, we get the condition for persistence of trophic level n :

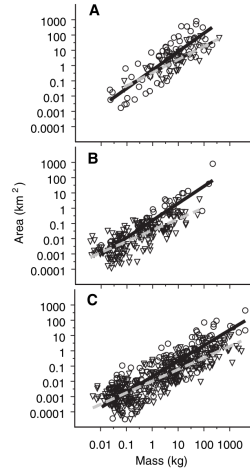
$$h_1 > \sum_{j=1}^n \frac{e_j}{c_j}$$

From which we can derive three conclusions

1. All things equal, occupancy should decrease with trophic level
2. Spatial dynamics imposes a limit to food chain length
3. Selection should promote larger colonization and lower extinction with increasing trophic level



Concept of spatial inefficiency Spatial use properties

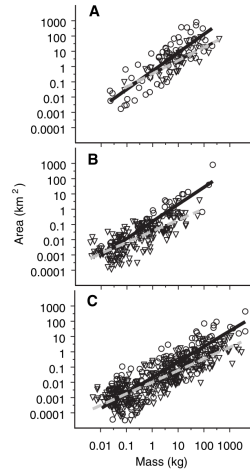


Jetz et al. 2004. Science



Concept of spatial inefficiency Spatial use properties

- ▶ Dispersal distance
- ▶ Foraging (home range)
- ▶ Migration distance

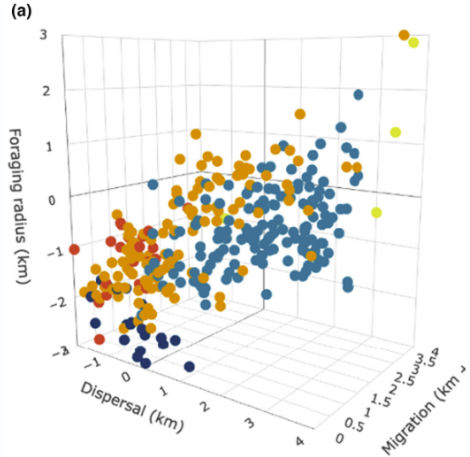


Jetz et al. 2004. Science



Concept of spatial inefficiency Spatial use properties

- ▶ Dispersal distance
- ▶ Foraging (home range)
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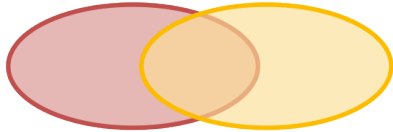
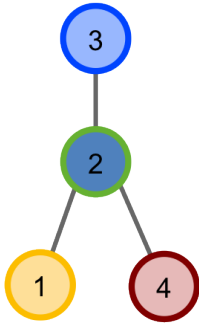


Straus et al. 2024. GEB.



Concept of spatial inefficiency

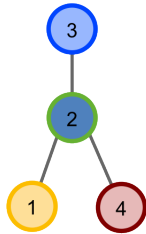
Diet breadth





Concept of spatial inefficiency

Diet breadth



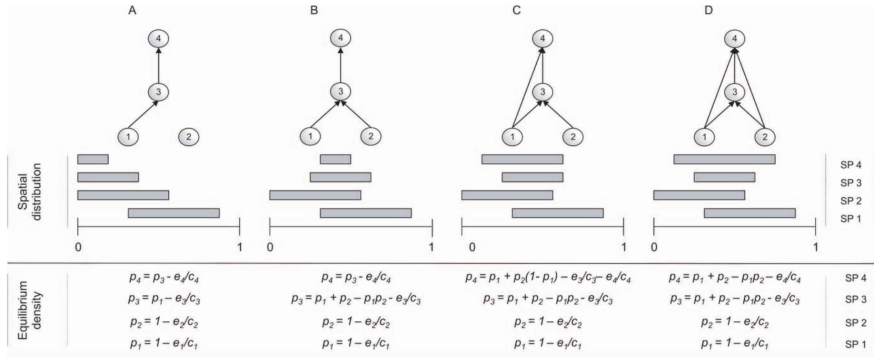
Attraction

Repulsion

Random



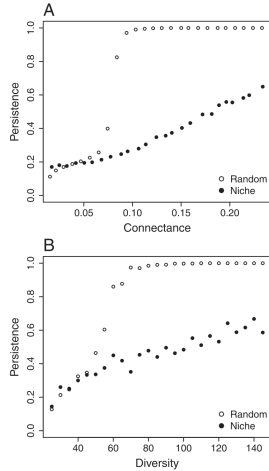
Concept of spatial inefficiency Summary



Gravel et al. 2011. PLOS ONE



Concept of spatial inefficiency Summary



Gravel et al. 2011. PLOS ONE

Exercise



Adding interactions to IBT

Now, using the regional food web of Havens from the github repository, explores the species-area relationships of the different trophic levels.

The basic assumptions remain the same :

1. A species can only invade an island if it has at least one prey present on the island
2. A species that loses its last prey on the island goes extinct

Implicit assumption : species with no prey in the metaweb are primary producers and therefore do not require prey to persist.

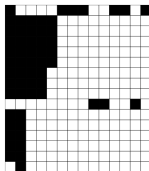
Trophic Island Biogeography



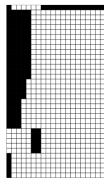
Island interaction networks



$S = 103, C = 0.07$



$S = 19, C = 0.16$



$S = 37, C = 0.1$

Comparison to the regional pool :

- ▶ Larger number of links
- ▶ Larger connectance
- ▶ Shorter food chains
- ▶ Fully connected network



Trophic IBT Model

Assumptions:

- ▶ A species can only invade an island if it has at least one prey present on the island
- ▶ A species that loses its last prey on the island goes extinct

$$\frac{dp_g}{dt} = c(1 - p_g)q_g - (e + \epsilon_g)p_g$$

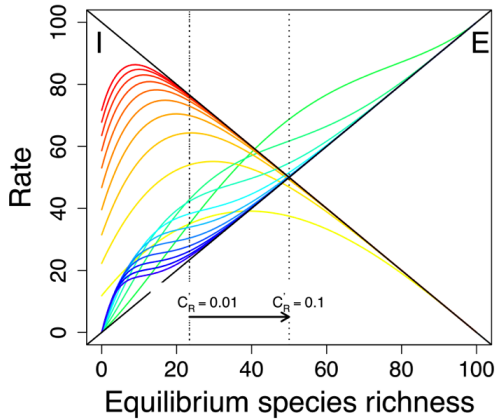
where:

$$p_g^* = \frac{cq_g}{cq_g + e + \epsilon_g}$$

- ▶ c : colonization rate
- ▶ e : community-independent extinction rate
- ▶ q_g : probability that a species has a prey present on the island
- ▶ ϵ_g : rate a predator loses its last prey on the island



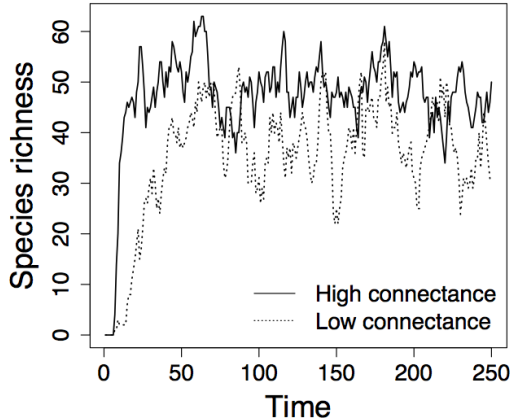
Equilibrium species richness



Gravel et al. (2011) *Ecology Letters*.



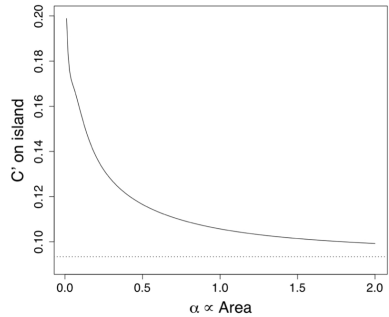
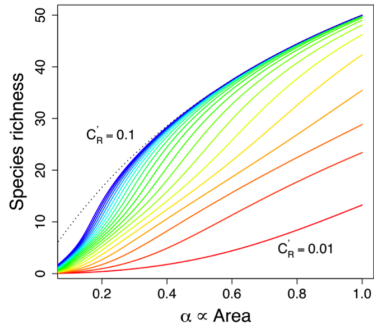
Network assembly dynamics



Gravel et al. (2011) *Ecology Letters*.

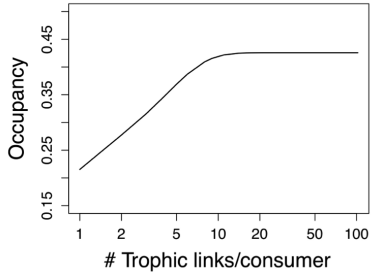


Local-regional feedback





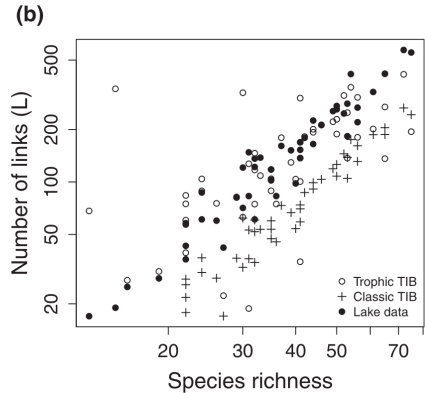
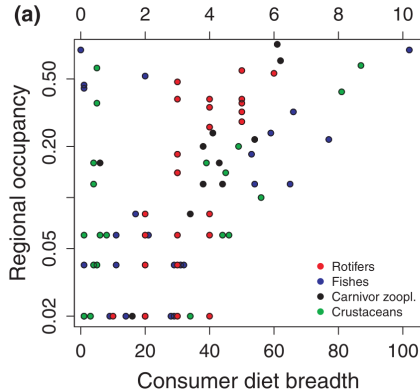
Testing theory Species distribution



Dataset	Classic TIB	Trophic TIB (analytical)	Trophic TIB (simulations)
Adirondack lakes	-2428.2	-2416.8	-2392.4
Florida islands	-259.3	-259.9	-260.0



Testing theory Scaling relationships

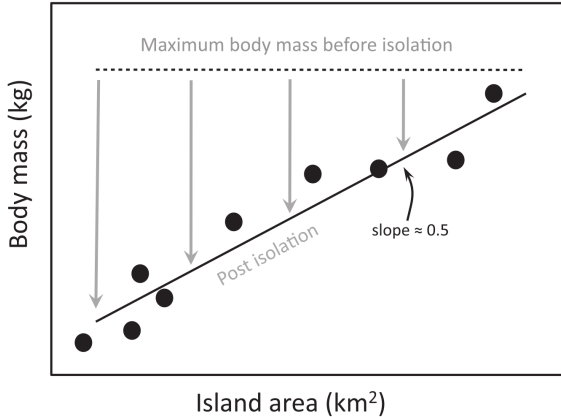


Gravel et al. (2011) *Ecology Letters*.

Trait-based IBT



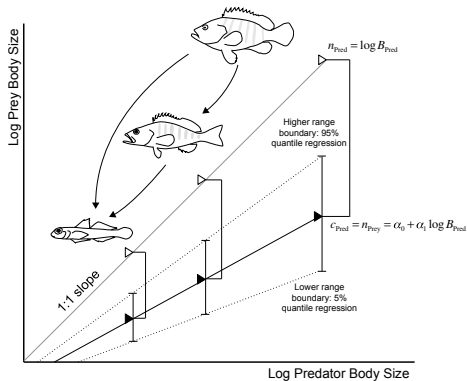
The mass-area relationship



Millien and Gonzalez (2011) J. Biogeo



Body-size and insular dynamics

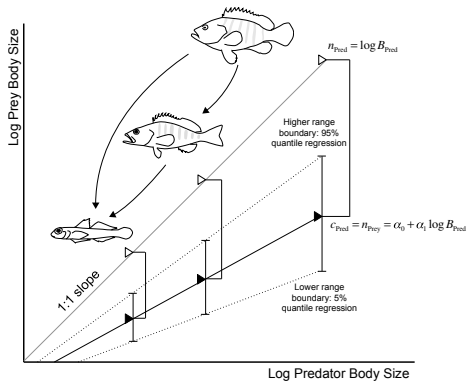


- ▶ Dispersal distance
- ▶ Number of propagules
- ▶ Energetic requirements
- ▶ Home range
- ▶ Population size
- ▶ Trophic level
- ▶ Diet breadth

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Body-size and insular dynamics

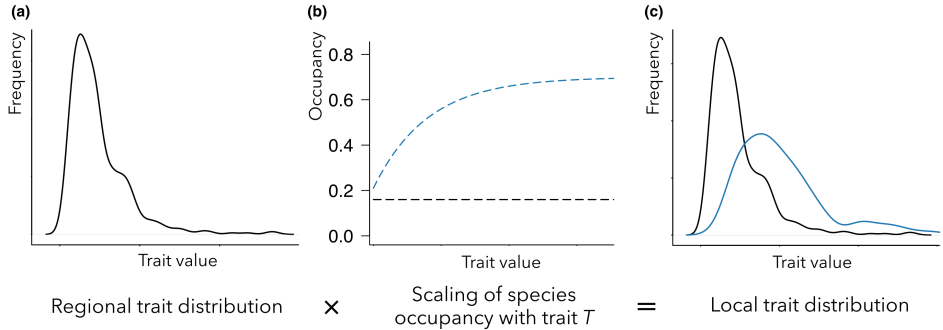


Additions to the model :

- Size-constraint food web
- Size-dependent c and e parameters



Islands as a non-random sample from the regional pool



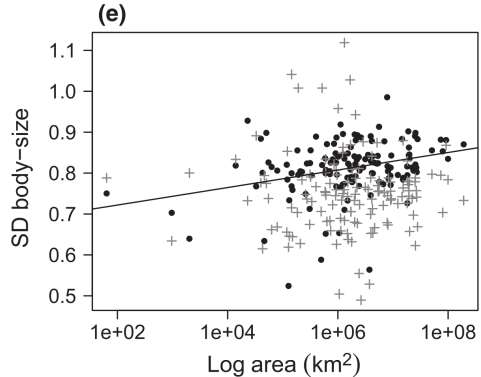
Jacquet et al (2017) Ecol. Lett.



Size distribution in coral reefs

Expectations :

- ▶ Variance in body-size increases (decreases) with area (isolation) for resources
- ▶ Variance in body-size decreases (decreases) with area (isolation) for consumers

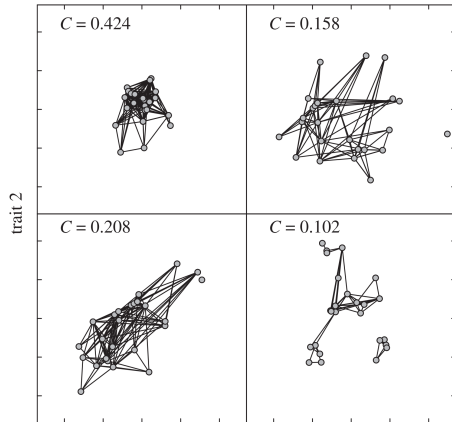


Jacquet et al (2017) Ecol. Lett.

And macro-evolution



Trait-matching concept

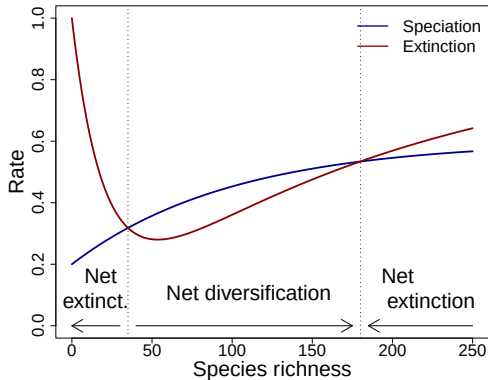


Gravel et al (2016) Phil. Trans. Roy. Soc.

- ▶ Network structure is the joint result of **trait-matching** and **trait-distribution**
- ▶ Traits could either evolve **neutrally** or under **biotic selection**



Trait evolution under biotic selection

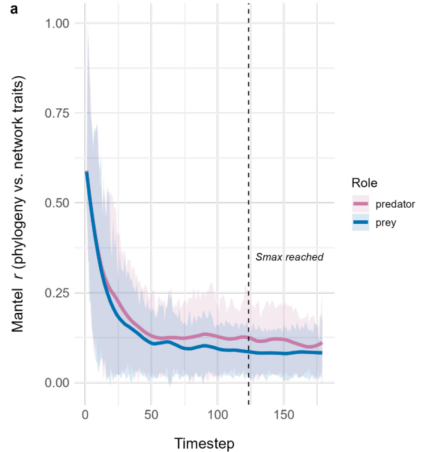
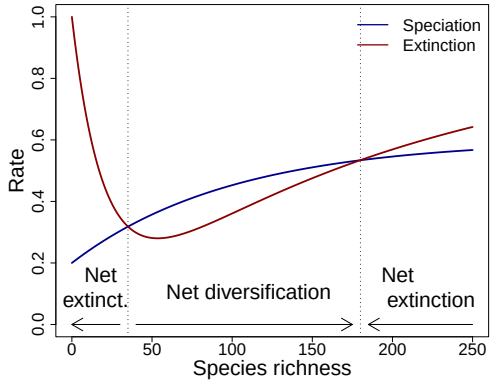


Additions to the model :

- ▶ Trait evolution at speciation
- ▶ Per species constant speciation rate
- ▶ Establishment contingent on prey availability
- ▶ Top-down effect on extinction
- ▶ Tracking of phylogeny, trait distribution and pairwise interactions



Trait evolution under biotic selection





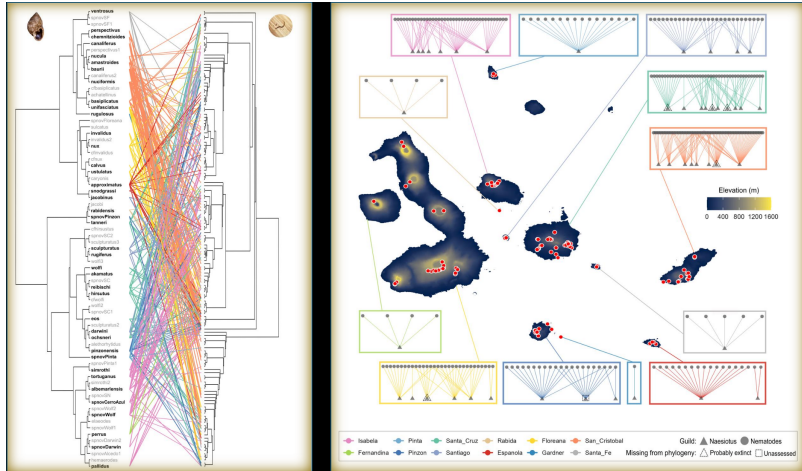
Phylogenetic signal in Galapagos food webs



- ▶ Regional metaweb approach
- ▶ 559 species, 96 birds, 40 reptiles, 20 mammals, 402 plants, 1 invertebrate
- ▶ Species checklists for 13 islands of various ages
- ▶ Interactions from the literature
- ▶ 2582 links, mean trophic rank of 3.55 (metaweb)



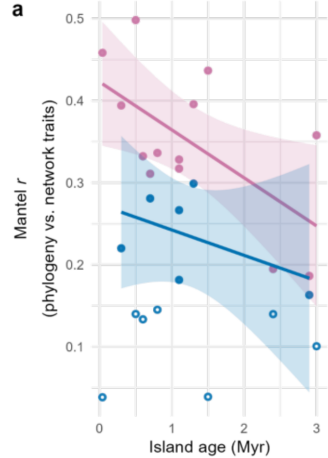
Phylogenetic signal in Galapagos food webs



Fuster et al. 2026. Am. Nat.



Phylogenetic signal in Galapagos food webs



Fuster et al. 2026. Am. Nat.

Conclusion



Conclusion

- ▶ Island biogeography theory can be expanded without sacrificing simplicity
- ▶ The key is to represent trait-dependence of colonization and extinction processes
- ▶ Local trait distributions are non-random samples of the regional pool
- ▶ Wide range of predictions, from species-area relationships to phylogenetic structure



Looking forward From functional anomalies to ecosystem functions

